

Elasticity

- Deforming force

↳ The external force applied on a body which causes change in shape and size of the body is called deforming force.

- Restoring Force

↳ The intermolecular force due to which a deformed body regains its original shape and size after the removal of deforming force is called restoring force.

- Elasticity

↳ The property of a body by virtue of which the body regain its original shape and size when the deforming force is removed is called elasticity.

- Plasticity

↳ The property of a body by virtue of which the body does not regain its original shape and size when the deforming force is removed is called plasticity.

• Stress

→ The restoring force per unit area of a deformed body is called stress.

$$\therefore \text{Stress} = \frac{\text{Restoring Force}}{\text{Area}} = \frac{F}{A}$$

In the elastic body, restoring force = deforming force

$$\therefore \text{Stress} = \frac{\text{Deforming force}}{\text{Area}}$$

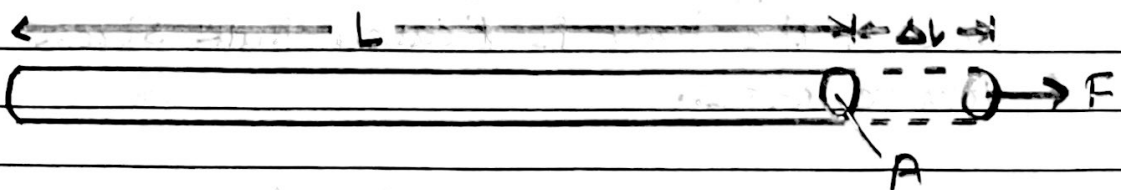
→ Types of stress

a) Normal stress

→ The stress developed in a body when deforming force is applied normally to the surface area is called normal stress.

$$\text{i.e. Normal Stress} = \frac{\text{Normal Force}}{\text{Area}} = \frac{F}{A}$$

Normal stress is responsible for the change in length of the body

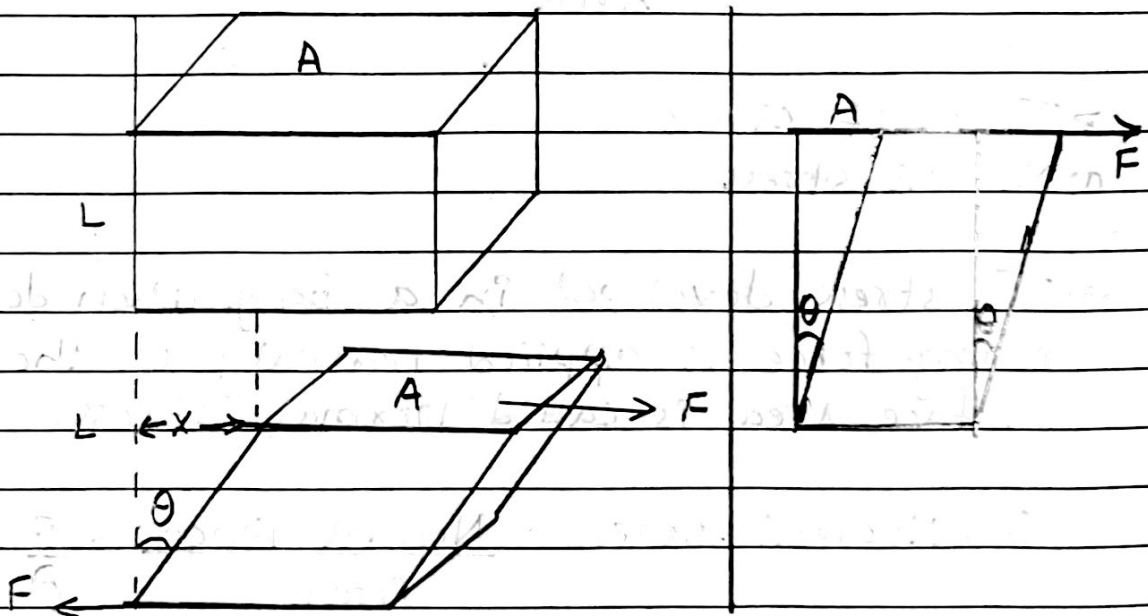


b) Tangential Stress or shearing stress:

↳ The stress developed in a body when deforming force is applied tangentially to the surface area is called tangential stress.

i.e. Tangential stress = $\frac{\text{Tangential Force}}{\text{Area}} = \frac{F}{A}$

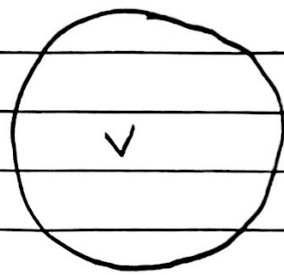
Tangential stress is responsible for the change in shape of body.



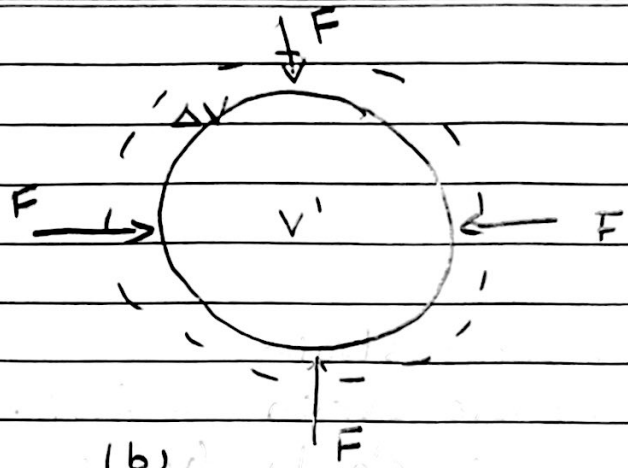
c) Bulk stress or Volume Stress

↳ The stress developed in a body when normal force is applied through the total surface area of the body is called bulk stress or Volume stress.

Volume stress is responsible for the change in volume of a body.



(a)



(b)

Fig: (a) Normal body & (b) Body with bulk stress.

• Strain

↳ The ratio of change in configuration (i.e. shape, length or Volume) to the original configuration of a deformed body is called strain.

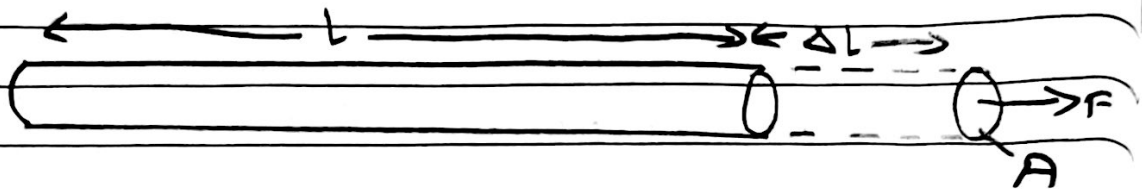
$$\text{i.e. Strain} = \frac{\text{Change in Configuration}}{\text{Original configuration}}$$

> Types of strain

i) Longitudinal Strain

↳ Longitudinal Strain is the ratio of change in length to the original length of a deformed body.

$$\text{i.e. Longitudinal Strain} = \frac{\text{change in length}}{\text{Original length}} = \frac{\Delta L}{L} \text{ or } \frac{e}{L}$$



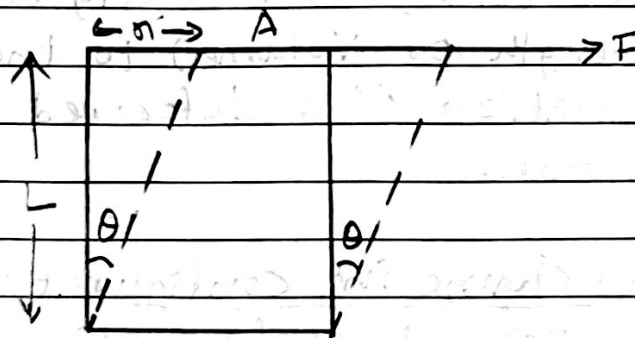
ii) Shear strain

→ Shear strain is defined as the angle through which the face of a body is turned when it is under the shearing stress.

i.e. $\tan \theta = \frac{\Delta x}{L}$

If θ is very small, $\tan \theta \approx \theta$

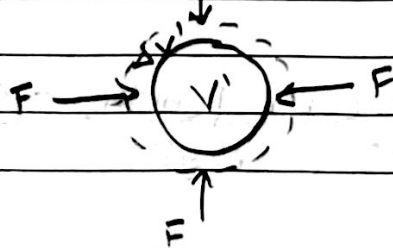
So, Shear strain, $\theta = \frac{\Delta x}{L}$



iii) Volumetric strain

→ Volumetric strain is the ratio of change in volume to the original volume of a deformed body.

i.e. $\text{Volumetric Strain} = \frac{\text{change in Volume}}{\text{Original Volume}} = \frac{\Delta V}{V}$



- Elastic Limit

↳ The maximum limit of deforming force up to which a body shows the property of elasticity is called elastic limit.

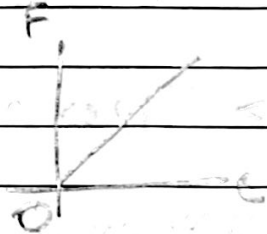
- Hooke's Law

↳ It states that "Within elastic limit, the restoring force acting on a body is directly proportional to its extension".

i.e. Restoring Force $\propto$ extension

or, $F \propto e$

$$\therefore F = ke \rightarrow (i)$$



Where, k is proportionality constant known as force constant.

Since,

Stress $\propto$ restoring force and strain $\propto$ extension

so, Hooke's law can also be written as,

Stress $\propto$ strain

Hence Hooke's law also states that the stress is directly proportional to the strain.

- Modulus of elasticity

According to Hooke's law,

Stress $\propto$ strain
or, Stress = $E \times$ strain

Where E is the proportionality constant called modulus of elasticity.

$$\therefore E = \frac{\text{stress}}{\text{strain}}$$

Hence modulus of elasticity is defined as the ratio of stress to the strain.

> Types of Modulus of Elasticity

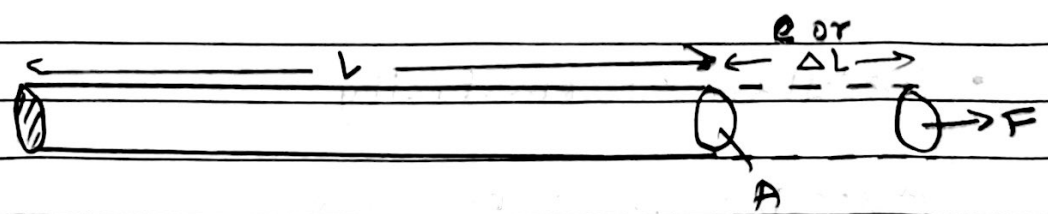
1) Young's Modulus of Elasticity (Y)

→ Young's modulus of elasticity is defined as the ratio of normal stress to the longitudinal strain.

i.e. Young's modulus (Y) = $\frac{\text{Normal stress}}{\text{Longitudinal strain}}$

$$\text{or, } Y = \frac{\frac{F}{A}}{\frac{e}{L}}$$

$$\therefore Y = \frac{FL}{Ae}$$



2) Shear Modulus or Modulus of Rigidity (η)

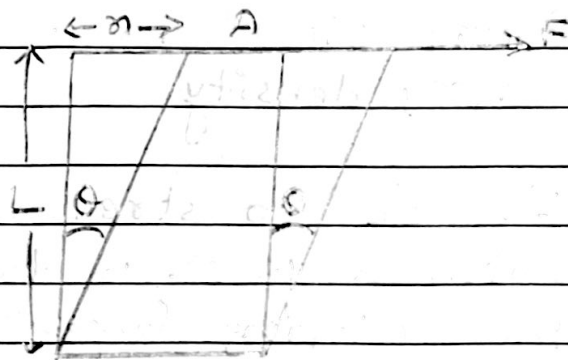
→ Modulus of rigidity is defined as the ratio of tangential stress to the shear strain within the elastic limit.

i.e. Modulus of rigidity, (η) = $\frac{\text{Tangential stress}}{\text{Shear strain}}$

$$\text{or, } \eta = \frac{F/A}{\theta}$$

$$\text{or, } \eta = \frac{\frac{F}{A}}{\frac{\Delta x}{L}}$$

$$\therefore \eta = \frac{FL}{A \cdot \Delta x}$$

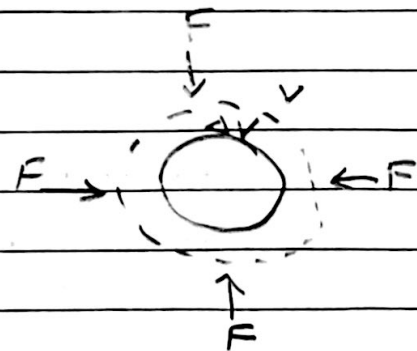


Where $\Delta x = x = \text{displacement}$

3) Bulk Modulus of Elasticity (B)

→ The ratio of Normal Stress to the Volume strain is called Bulk Modulus of Elasticity.

$$\begin{aligned} \therefore B &= \frac{\text{Normal Stress}}{\text{Volume strain}} \\ &= \frac{\frac{F}{A}}{\frac{\Delta V}{V}} \end{aligned}$$



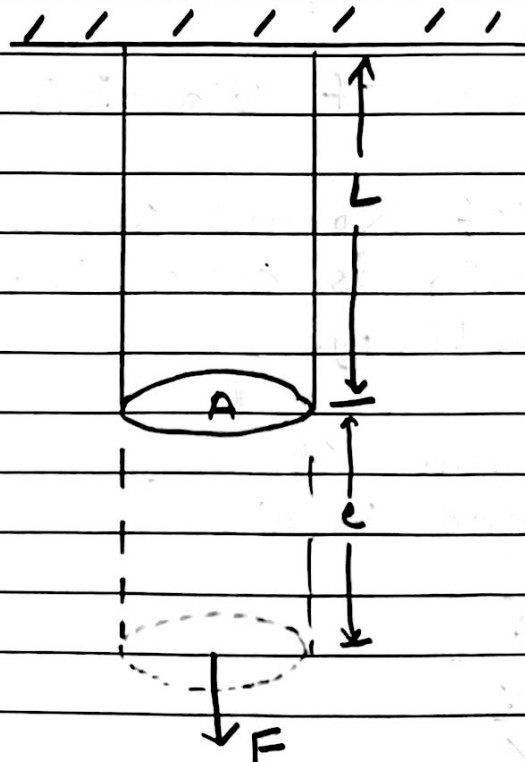
$$= - \frac{P}{\frac{\Delta V}{V}} \quad \therefore P = \frac{F}{A} \text{ is pressure}$$

$$= - \frac{P \Delta V}{\Delta V}$$

Where -ve sign indicates that when pressure increases then volume decreases and vice versa.

✓✓ Energy Stored in a stretched wire and energy density

In order to stretch a wire, some external work is to be done against the restoring force. This work done stores in the wire as its potential energy.



Let e be the extension produced in a wire of length L and cross-sectional Area (A) when normal force F is applied to it.

Now,

Young's modulus of elasticity of the ~~ext~~ curve is given by:

$$Y = \frac{\text{Normal stress}}{\text{Longitudinal strain}}$$

$$\text{or, } Y = \frac{\frac{F}{A}}{\frac{e}{L}}$$

$$\text{or, } Y = \frac{FL}{Ae}$$

$$\therefore F = \frac{Y Ae}{L} \rightarrow (1)$$

Let, de be the small extension of wire due to this force then small work done in stretching the wire is given by

$$dW = F de$$

$$\therefore dW = \frac{Y Ae}{L} de \rightarrow (2)$$

Now,

Total work done in stretching the wire is given by,

$$W = \frac{YA}{L} \int_0^e e \, de$$

$$= \frac{YA}{L} \left[\frac{e^2}{2} \right]_0^e$$

$$= \frac{YA}{2L} [e^2 - 0]$$

$$= \frac{Y A e^2}{2L}$$

$$= \frac{1}{2} \times \frac{Y A e}{L} \times e$$

$$= \frac{1}{2} \times F \times e \quad \therefore F = \frac{Y A e}{L}$$

→ ③

This work done stores in the wire as its elastic potential energy. So, energy stored in the stretched wire is given by

$$E = \frac{1}{2} F \times e \rightarrow \textcircled{4}$$

This shows that the energy stored in a stretched wire is equal to half of product of force and extension.

• Compressibility

→ The reciprocal of bulk modulus of elasticity is called compressibility.

$$\text{i.e. } C = \frac{1}{B}$$

• Energy density

→ The energy stored per unit volume of a stretched wire is called its energy density.

i.e. Energy density (U) = $\frac{\text{Energy stored}}{\text{Volume}}$

$$= \frac{1}{2} F x e$$

$$A \times L$$

$$= \frac{1}{2} \times \frac{F}{A} \times \frac{e}{L}$$

$$= \frac{1}{2} \times \text{Stress} \times \text{Strain} \rightarrow (5)$$

This shows that the energy density of a stretched wire is equal to the half of product of stress and strain.

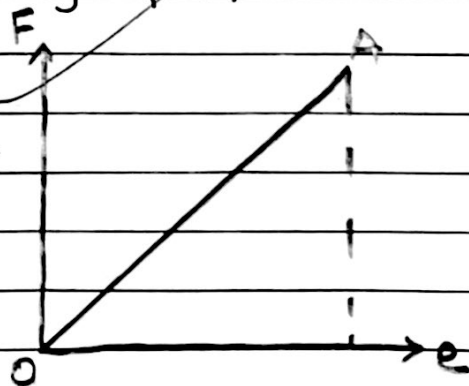
Note:

In Force - extension graph,

- ① Slope of line OA gives force constant (k)

$$\text{Slope} = \frac{F}{e}$$

$$= k \quad [F = k e]$$



② Area under the line OA gives energy stored

$$\therefore A = \frac{1}{2} \epsilon \times F$$

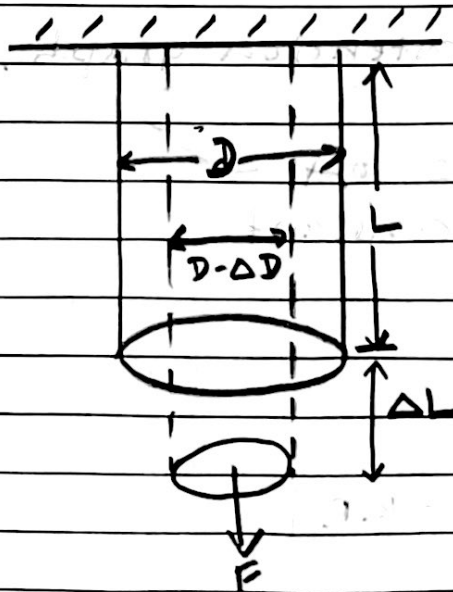
= Energy stored.

• Poisson's ratio

→ The ratio of Lateral strain to the Longitudinal Strain is called Poisson's ratio.

$\therefore$ Poisson's ratio = $\frac{\text{Lateral strain}}{\text{Longitudinal Strain}}$

i.e. $\sigma = \frac{\beta}{\alpha} \rightarrow (i)$



Here,

Lateral strain (β) = $\frac{\text{change in diameter}}{\text{original diameter}}$

$$= \frac{\Delta D}{D}$$

And,

Longitudinal strain (α) = $\frac{\text{change in length}}{\text{original length}}$

$$= \frac{\Delta L}{L}$$

Then,

From (1)

$$\sigma = \frac{\Delta D}{D} \times \frac{\Delta L}{L}$$

$$\therefore \sigma = \frac{\Delta D}{D} \times \frac{L}{\Delta L} \rightarrow (2)$$

• Elastic Fatigue

→ The property of elastic body by virtue of which the body loses its elastic behaviour under the action of repeated loading and unloading for a long time is called elastic fatigue.

Homework

- 1) Will the young's modulus of elasticity change if the load hanging on it is doubled? why?

Ans: The young's modulus of elasticity of a body is given by

$$Y = \frac{F/A}{e/l} = \frac{Fl}{Ae}$$

Where, $\frac{F}{A}$ = stress, $\frac{e}{l}$ = strain

When the load hanging on the material is doubled, the Young's modulus of elasticity remains constant because the increase in load is balanced by the extension produced. The young's modulus of elasticity of a material is constant quantity. It depends on the nature of material, but not on what stress is applied to it. Therefore, Young's modulus of elasticity remains same although the load hanging on it is changed.

- ⑤ A resistance band is a length of an elastic material that can be used for exercise.

The user repeatedly applies an increasing tensile force and then releases the force. The force extension graph for resistance band is shown. Describe with reference to the graph, the behaviour of the resistance band when it is repeatedly loaded and unloaded.

④ Here, ^{Stored}
Energy stored during loading = Area
under the loading curve
and

Energy ~~is~~ released during unloading
= Area under the unloading curve.

Here, energy ^{stored} : > Energy released

So,

Energy absorbed = Energy stored - Energy
released

= Area of hysteresis loop

The absorbed Energy converts into heat.
So. the band gets warm during use.

2) Explain why steel is said to be more elastic than rubber.

→ Steel is said to be more elastic than rubber because when same force is applied to steel and ~~the~~ rubber of equal length and cross section area, we find out that the extension in the rubber is much greater than extension in steel. Therefore, for a given stress, the strain produced in the ~~rubber~~ steel is much more smaller than in a rubber. This implies that young's modulus for steel is greater than that for rubber. Therefore, steel is more elastic than rubber.

3) Two wires A and B have equal lengths and are made of same material. If the diameter of wire A is twice that of wire B, which wire has the greater extension for a given load?

Ans Wire B will have the greater extension. Since wire A has twice the diameter, it has four times the cross-sectional area ($A = \pi r^2$). Extension is inversely proportional to area ($\Delta L \propto \frac{1}{A}$). Therefore, under the same load, wire B (the thinner one) will stretch four times as much as wire A.

4) Why do spring balances show wrong reading after long use?

Ans Spring show wrong reading due to elastic fatigue. After long-term use or being overloaded, the material loses its ability to return to its original length (it passes its elastic limit). This permanent deformation results in a "zero error" or inaccurate scaling.

• Numerical

- 1) A wire of length 2.5 m and area of cross-section $1 \times 10^{-6} \text{ m}^2$ has a mass of 15 kg hanging on it. What is the extension produced? How much is the energy stored in the extended wire if Young's modulus of wire is $2 \times 10^{11} \text{ Nm}^{-2}$.

→ Soln.

$$A = 1 \times 10^{-6} \text{ m}^2 \quad l = 2.5 \text{ m}$$

$$m = 15 \text{ kg}$$

$$e = ?$$

$$E = ?$$

$$Y = 2 \times 10^{11} \text{ Nm}^{-2}$$

Now,

$$Y = \frac{F l}{A e}$$

$$2 \times 10^{11} = \frac{m g \times 2.5}{1 \times 10^{-6} \times e}$$

$$\text{or, } e = \frac{15 \times 10 \times 2.5}{1 \times 10^{-6} \times 2 \times 10^{11}}$$

$$\text{or, } e = 1.875 \times 10^{-3} \text{ m}$$

Again,

$$E = \frac{1}{2} e \cdot X F$$

$$= \frac{1}{2} \times 1.875 \times 10^{-3} \times 15 \times 10$$

$$= 0.143 \text{ J}$$

- 2) A uniform steel wire of density $8,000 \text{ kg m}^{-3}$ weight 20 g and is 2.5 m long. It lengthens by 1 mm when stretched by a force of 80 N . Calculate the value of Young's modulus of steel and the energy stored in the wire.

→ soln,

$$e = 1 \text{ mm} = 1 \times 10^{-3} \text{ m} \quad l = 2.5 \text{ m}$$

$$m = 20 \text{ g} = 0.02 \text{ kg}$$

$$\rho = 8000 \text{ kg m}^{-3}$$

$$F = 80 \text{ N}$$

Now,

$$V = \frac{m}{\rho} = \frac{0.02}{8000} = 2.5 \times 10^{-6} \text{ m}^3$$

$$A = \frac{V}{l} = \frac{2.5 \times 10^{-6}}{2.5} = 1 \times 10^{-6} \text{ m}^2$$

~~last~~ Again,

$$Y = \frac{Fl}{Ae}$$

$$Y = \frac{80 \times 2.5}{1 \times 10^{-6} \times 1 \times 10^{-3}}$$

$$Y = \frac{80 \times 2.5}{10^{-9}}$$

$$Y = 2 \times 10^{11} \text{ N m}^{-2}$$

Last,

$$E = \frac{1}{2} Fxe$$

$$= \frac{1}{2} \times 80 \times 10^{-3}$$

$$= 0.04 \text{ J}$$

5: A ~~wire stretches~~ rubber cord of a catapult has a cross-sectional area 1 mm^2 and total unstretched length 10.0 cm . It is stretched to 12.0 cm and then released to project a missile of mass 5.0 gm . Calculate the velocity of projection of ~~projec~~ missile. (γ for rubber $= 5.0 \times 10^8 \text{ N m}^{-2}$).

→ Soln,

$$A = 1 \text{ mm}^2 = 1 \times 10^{-6} \text{ m}^2$$

$$L = 10 \text{ cm} = 10 \times 10^{-2} \text{ m}$$

$$e = (12 - 10) = 2 \text{ cm} = 2 \times 10^{-2} \text{ m}$$

$$m = 5 \text{ gm} = 5 \times 10^{-3} \text{ kg}$$

$$\gamma = 5 \times 10^8 \text{ N/m}^2$$

$$V = ?$$

Here,

K.E. of missile = Energy stored in rubber.

$$\text{or, } \frac{1}{2} m v^2 = \frac{1}{2} F x e$$

$$\text{or, } m v^2 = F x e$$

$$\text{or, } m v^2 = \frac{\gamma A e}{L} x e \left[\because \gamma = \frac{F L}{A e} \right]$$

$$\text{or, } 5 \times 10^{-3} \times v^2 = \frac{5 \times 10^8 \times 1 \times 10^{-6} \times (2 \times 10^{-2})^2}{10 \times 10^{-2} \text{ m}}$$

$$\text{or, } v^2 = \frac{0.2}{5 \times 10^{-4}}$$

$$\text{or, } v = \sqrt{400}$$

$$\therefore v = 20 \text{ m/s}$$

6) A specimen of oil having initial volume of 600 cm^3 is subjected to pressure increase of $3.6 \times 10^6 \text{ Pa}$ and volume is found to decrease by 0.45 cm^3 . What is the Bulk modulus and compressibility of material?

→ Soln,

$$V = 600 \text{ cm}^3$$

$$P = 3.6 \times 10^6 \text{ Pa}$$

$$\Delta V = 0.45 \text{ cm}^3$$

$$B = ?$$

$$C = ?$$

Now,

$$B = \frac{F}{A}$$

$$\frac{\Delta V}{V}$$

$$= \frac{P}{\frac{\Delta V}{V}} \quad \left[\because P = \frac{F}{A} \right]$$

$$= \frac{PV}{\Delta V}$$

$$= \frac{3.6 \times 10^6 \times 600}{0.45}$$

$$= 4.8 \times 10^9 \text{ Pa}$$

Again,

$$C = \frac{1}{B} = \frac{1}{4.8 \times 10^9}$$

$$= 2.08 \times 10^{-10} / \text{Pa}$$

3) A vertical brass rod of circular section is loaded by placing a 5 kg weight on top of it. If its length is 50 cm and radius of cross section is 1 cm, find the contraction of rod and the energy stored in it. Given γ for brass = 3.5×10^{10} N/m.

→ soln,

$$m = 5 \text{ kg}$$

$$L = 50 \text{ cm} = 0.5 \text{ m}$$

$$r = 1 \text{ cm} = 0.01 \text{ m}$$

$$\gamma = 3.5 \times 10^{10} \text{ N/m}$$

$$F = mg = 5 \times 10 = 50$$

$$A = \pi (1 \text{ cm})^2 = \pi (0.01 \text{ m})^2 = 3.14 \times 10^{-4}$$

$$\gamma = \frac{FL}{Ae}$$

$$\text{or, } 3.5 \times 10^{10} = \frac{50 \times 0.5}{3.14 \times 10^{-4} e}$$

$$\text{or, } 3.5 \times 10^{10} = \frac{25}{3.14 \times 10^{-4} e}$$

$$\text{or, } 3.5 \times 10^{10} = \frac{79617.83}{e}$$

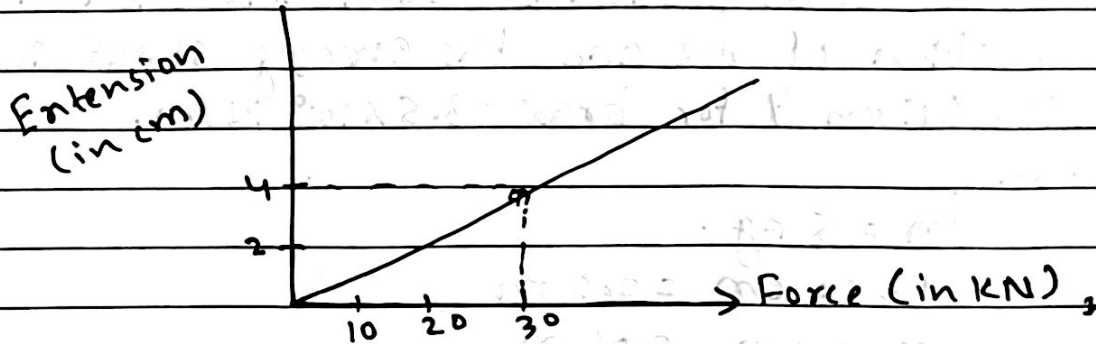
$$\therefore e = 2.27 \times 10^{-6} \text{ m}$$

$$\text{Energy stored} = \frac{1}{2} \times e \times F$$

$$= \frac{1}{2} \times 2.27 \times 10^{-6} \times 50$$

$$= 5.68 \times 10^{-5} \text{ J}$$

4) The graph shows how the extension of a spring varies with the force used to stretch it.



i) Determine the force constant of the spring.

Given, $F = 30 \text{ kN} = 30 \times 10^3 \text{ N}$
 $e = 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$

Now,

$$k = \frac{F}{e} = \frac{30 \times 10^3 \text{ N}}{4 \times 10^{-2} \text{ m}}$$

$$= 7.5 \times 10^5 \text{ N m}^{-1}$$

ii) What is the energy stored in the spring when the extension is 4 cm?

Energy stored $= \frac{1}{2} \times F \times e$

$$= \frac{1}{2} \times 30 \times 10^3 \times 0.04$$

$$= 600 \text{ J}$$